

CSE 4108

☰ Instructor	
👤 Class Note of	
📅 Exam Date	
# ID	
☰ Subject	Structured Programming I Lab
⋮ tag	

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LAB – 2 16-8-22

Task – 1

Printing Tick: Write a program that uses printf to display the following picture on the screen:

```

      *
     *
    *
   *
  * *
 * *
*

```

```
#include<stdio.h>

int main ()

{
    printf("      *\n");
    printf("     *\n");
    printf("    *\n");
    printf("   *\n");
    printf("  * *\n");
    printf(" * *\n");
    printf("*\n");

    return 0;
}
```

Task - 2

The volume of a sphere is calculated by: $(4/3)\pi r^3$ where r is the radius of the sphere.

Write a C program that computes the volume of a sphere with a 20-meter diameter. Use $\pi = 3.14159$.

```
#include<stdio.h>

int main()

{
    float diamiter, r,pi,v;

    diamiter = 20;
    pi = 3.14159;
    r = diamiter / 2 ;

    v = (4/3)*pi*r*r*r;

    printf ("Volume is %.2f",v);

    return 0;
}
```

Task - 3

Modify the program written in the last task so that it prompts the user to enter the radius of the sphere.

```
#include<stdio.h>

int main()

{
    float r,pi,v;

    pi = 3.14159;

    printf("input the radius\t:");

    scanf("%f",&r);
}
```

```

v = (4.0f/3.0f)*pi*r*r*r;

printf ("Volume of the sphere is is %f",v);

return 0;

}

```

Task - 4

Bvaly is a multinational business venture that focuses on e-commerce. To cover the maintenance cost, they charge 5% VAT for any item purchased. For example: If you buy a product worth \$100.00, you have to pay \$105.00 (including VAT). They have hired you to automate the process of calculating the amount a customer needs to pay. Write a C program that asks the user to enter the dollar-and-cents amount, then displays the amount with 5% VAT added.

```

#include<stdio.h>

int main ()

{
    float val,vat,tot;

    printf ("\n\n\tPlease input the amount :\t");

    scanf ("%f",&val);

    vat = (val/100)*5;

    tot = val + vat;

    printf("\n\n\tcustomer have to pay \t %.2f $\n\n",tot);

    return 0;

}

```

Task - 5

Write a C program that asks the user to enter the value of x and outputs the value of the following polynomial:

$$3x^5 + 2x^4 - 5x^3 - x^2 + 6x - 6$$

```

#include<stdio.h>>

int main ()

{
    float x, ans;

    printf ("please input the value of the x :\t");
    scanf ("%f",&x);

    ans = (3.0f*x*x*x*x*x) + (2.0f*x*x*x*x) - (5.0f*x*x*x) - (x*x) + (7.0*x) - 6 ;

    printf("the answer is = %f",ans);

    return 0;

}

```

Task - 6

Modify the program of the last task so that the polynomial is evaluated using the following formula:

$$(((3x + 2)x - 5)x - 1)x + 7)x - 6$$

```
#include<stdio.h>
#include<math.h>

int main ()

{
    float x, ans;

    printf ("please input the value of the x :\t");
    scanf ("%f",&x);

    ans = (((((3*x)+2)*x-5)*x+1)*x+7)*x-6;

    printf("the answer is = %f",ans);

    return 0;
}
```

Task - 7

Convenience Delivery Store(CDS) of Imperial University of Tokyo (IUT) is a retail business that stocks a range of everyday items such as snack foods, confectionery, soft drinks, toiletries, etc. As the university houses international students, the store accepts \$ as payment currency. Everyday numerous students get their daily goods from CDS. After purchasing an item, whenever a student gives a note of large denomination (like \$500 for example), the owner of the shop Mr. Sakamoto faces a huge problem. He can easily calculate the amount to be returned.

But he doesn't know how to pay that amount using the smallest number of denominations.

Mr. Sakamoto knows you're a great programmer and asked for your help. You need to write a program that takes the U.S. dollar amount as input and then shows how to pay that amount using the smallest number of \$20, \$10, \$5, and \$1 bills.

Mr. Sakamoto has assured you that he will not enter any fractional value as input.

```
#include <stdio.h>

int main()

{
    int i,twn,rem_1,ten,rem_2,fiv,rem_3;

    printf("\n\n\t\tMr Sakamoto, input the ammount please :\t\t\t");

    scanf ("%d",&i);

    twm = i/20;

    printf("\n\n\n\t\t\t20$ bills :\t %d", twm);

    rem_1 = i%20;

    ten = rem_1/10;

    printf("\n\n\n\t\t\t10$ bills :\t %d", ten);

    rem_2 = rem_1%10;

    fiv = rem_2/5;

    printf("\n\n\n\t\t\t5$ bills :\t %d", fiv);

    rem_3 = rem_2%5;

    printf ("\n\n\n\t\t\t1$ bills :\t %d",rem_3);

    return 0;

}
```

Task - 8

ABC Bank Ltd. provides loans to its customers. The customers return the amount by a fixed monthly payment. On each month however, the remaining balance is incremented by the interest rate assigned for each customer. The total balance at the end of each month can be calculated using the

following formula: $\text{currBalance} = \text{prevBalance} - \text{pay} + (\text{prevBalance} \times \text{rate})$

Here,

currBalance = Updated Balance

prevBalance = Previous Balance

pay = Monthly Payment

rate = Monthly Interest Rate

Your task is to write a program that calculates the remaining payable amount on a loan after the first, second, and third monthly payment.

Your program will take the loan amount, yearly interest rate, fixed monthly payment as input. You need to display each balance with two digits after the decimal point.

```
#include<stdio.h>

int main ()
{
    float cb_1, prevBalance, pay, rate, rrate, cb_2, cb_3;

    //taking inputs

    printf("\n\nplease input the amount of loan : \t");
    scanf ("%f",&prevBalance);

    printf ("\n\nplease input your interest rate : \t");
    scanf ("%f",&rrate);

    printf ("\n\nplease input your monthly payment : \t");
    scanf ("%f",&pay);

    //defining rate

    rate = rrate / (100*12) ;

    //the final calculation of loan
    // cb_1 stands for current balance for 1 month and so on.

    cb_1 = prevBalance + (prevBalance*rate) - pay;
    printf ("\n\nBalance remaining after first payment: \t %f",cb_1);

    cb_2 = cb_1 + (cb_1*rate) - pay;
    printf ("\n\nBalance remaining after second payment: \t %f",cb_2);

    cb_3 = cb_2 + (cb_2*rate) - pay;
    printf ("\n\nBalance remaining after third payment: \t %f\n\n",cb_3);

    return 0;
}
```

LAB - 2 23/8/22

Task - 1

It's a Date: Write a program that accepts a date from the user in the format mm/dd/yyyy and then displays it in the form yyyymmdd.

Sample Run:

Enter a date (mm/dd/yyyy): 2/17/2021

You entered the data 20210217

```
#include<stdio.h>

int main ()

{
    int num,sum, d1,rem1, d2, rem2, d3, rem3, d4, rem4;

    printf ("\n\n\tEnter a five-digit integer : ");
    scanf ("%d",&num);

    d1 = num/10000;

    rem1 = num%10000;

    d2 = rem1/1000;

    rem2 = rem1%1000;

    d3 = rem2/100;

    rem3 = rem2%100;

    d4 = rem3/10;

    rem4 = rem3%10;

    sum = d1+d2+d3+d4+rem4;

    printf("\n\n\tThe sum is : %d\n\n", sum);

    return 0;
}
```

Task - 2

Inventory Management System: Mr. Sakamoto from Convenience Delivery Store (CDS) of Imperial University of

Tokyo (IUT) is back. This time, he wants you to create a list of items in his shop. He will provide you item number, unit price and purchase date of the products available in CDS. He wants you to show the items in an organized manner. However, before finalizing the design he wants to see how the list will look like a single item. You must write a program that formats product information entered by the user. A session

```
#include<stdio.h>

int main ()

{
    int item_num, date, month, yr;
    float price;

    printf("Enter item number : ");
    scanf("%d",&item_num);

    printf("\n\nEnter unit price : ");
    scanf("%f",&price);

    printf("\n\nEnter purchase date (mm/dd/yyyy) : ");
    scanf("%d/%d/%d",&month,&date,&yr);

    printf ("\n\nItem price\tDate\t\tUnit Purchase");
    printf (" \n\n$ %3.2f\t\t%.2d/%.2d/%d\t%13d\n\n",price,month,date,yr,item_num);

    return 0;
}
```

Task - 3

Books are identified by an International Standard Book Number (ISBN). ISBNs assigned after January 1, 2007 contain 13 digits, arranged in five groups, such as 978-0393-97950-3. (Older ISBNs use 10 digits). The first group (the GS1 prefix) is currently either 978 or 979. The group identifier specifies the language or country of origin (for example, 0 and 1 are used in English speaking countries). The publisher code identifies the publisher. The item number is assigned by the publisher to identify a specific book. An ISBN ends with a check digit that's used to verify the accuracy of the preceding digits. Write a program that breaks down an ISBN entered by the user.

```
#include<stdio.h>

int main ()

{
    int gs1,GI,PC,In,cd;

    printf ("Enter ISBN : ");
    scanf ("%d-%d-%d-%d",&gs1,&GI,&PC,&In,&cd);

    printf("\n\nGS1 prefix :      %d",gs1);
    printf("\n\nGroup Identifier :  %d",GI);
    printf("\n\nPublisher Code :    %d",PC);
    printf("\n\nItem Number :        %d",In);
    printf("\n\nCheck Digit :         %d\n\n",cd);

    return 0;
}
```

Task - 4

Write a program that prompts the user to enter a telephone number in the form (xxx)xxx-xxxx and then displays the number in the form xxx.xxx.xxxx.

Sample Run:

Enter phone number [(xxx) xxx-xxxx]: (404)817 - 6900

You entered 404.817.6900

```
#include<stdio.h>

int main ()

{
    int a,b,c;

    printf("enter phone number : [(xxx)xxx-xxxx] : ");
    scanf ("%d)%d-%d",&a,&b,&c);

    printf ("\n\nYou entered %d.%d.%d\n\n",a,b,c);

    return 0;
}
```

Task - 5

Write a program that asks the user to enter the numbers from 1 to 16 (in any order) and then displays the numbers in a 4 by 4 arrangement, followed by the sums of the rows, columns, and diagonals.

```
#include<stdio.h>

int main ()

{
    int a1,a2,a3,a4,a5,a6,a7,a8,a9,a10,a11,a12,a13,a14,a15,a16,rs1,rs2,rs3,rs4;
    int c1,c2,c3,c4,d1,d2;

    printf("Enter the numbers from 1 to 16 in any order : \n\n");

    scanf ("%d %d %d %d %d %d %d %d %d %d %d %d %d %d %d",&a1,&a2,&a3,&a4,&a5,&a6,&a7,&a8,&a9,&a10,&a11,&a12,&a13,&a14,&a15,&a16);

    printf("\n\t%d\t%d\t%d\t%d",a1,a2,a3,a4);
    printf("\n\t%d\t%d\t%d\t%d",a5,a6,a7,a8);
    printf("\n\t%d\t%d\t%d\t%d",a9,a10,a11,a12);
    printf("\n\t%d\t%d\t%d\t%d",a13,a14,a15,a16);

    rs1 = a1+a2+a3+a4;
    rs2 = a5+a6+a7+a8;
    rs3 = a9+a10+a11+a12;
    rs4 = a13+a14+a15+a16;

    c1 = a1+a5+a9+a13;
    c2 = a2+a6+a10+a14;
    c3 = a3+a7+a11+a15;
    c4 = a4+a8+a12+a16;

    d1 = a1+a6+a11+a16;
    d2 = a4+a7+a10+a13;

    printf ("\n\nRow sums : %d %d %d %d",rs1,rs2,rs3,rs4);
    printf ("\nColumn sums : %d %d %d %d",c1,c2,c3,c4);
    printf ("\ndiagonal sums : %d %d\n\n",d1,d2);

    return 0;
}
```

Task - 6

Consider a C program that prompts the user to enter two fractions like the given sample input and then displays their sum. Fractions are customarily denoted as: numerator/denominator.

```
#include<stdio.h>
```



```

int main ()

{
    int n1,d1,n2,d2,n,d;

    printf ("\tEnter two fractions separated by a plus sign : ");
    scanf( "%d/%d+%d/%d",&n1,&d1,&n2,&d2);

    d = d1*d2;
    n = (n1*d2)+(n2*d1);

    printf("\n\n\tThe sum is : %d/%d\n\n",n,d);

    return 0;
}

```

Task - 7

Given a five-digit integer, print the sum of its digits.

```

#include<stdio.h>

int main ()

{
    int num,sum, d1,rem1, d2, rem2, d3, rem3, d4, rem4;

    printf ("\n\n\tEnter a five-digit integer : ");
    scanf( "%d",&num);

    d1 = num/10000;

    rem1 = num%10000;

    d2 = rem1/1000;

    rem2 = rem1%1000;

    d3 = rem2/100;

    rem3 = rem2%100;

    d4 = rem3/10;

    rem4 = rem3%10;

    sum = d1+d2+d3+d4+rem4;

    printf("\n\n\tThe sum is : %d\n\n", sum);

    return 0;
}

```

LAB - 4 30/8/22

Task - 1

Write a program that asks the user to enter a two-digit number, then prints the number with its digits reversed.

```

#include<stdio.h>

int main()

{

```

```

int num,n1,n2;

printf ("Enter a two-digit number : ");
scanf ("%d",&num);

n1 = num/10;
n2 = num%10;

printf ("The reversal is : %d%d\n\n",n2,n1);

return 0;
}

```

Task - 2

Extend the program from the previous task to handle three-digit numbers.

```

#include<stdio.h>

int main()

{
    int num,n1,n2,n3,rem1;

    printf ("Enter a three-digit number : ");
    scanf ("%d",&num);

    n1 = num/100;
    rem1 = num%100;
    n2 = rem1/10;
    n3 = rem1%10;

    printf ("The reversal is : %d%d%d\n\n",n3,n2,n1);

    return 0;
}

```

Task - 3

Rewrite the program from the previous task so that it prints the reversal of a three-digit number without using arithmetic to split the number into digits.

```

#include<stdio.h>

int main()

{
    char a,b,c;

    printf ("Enter a three-digit number : ");

    scanf ("%c%c%c",&a,&b,&c);

    printf("The reversal is : %c%c%c",c,b,a);

    return 0;
}

```

Task - 4

Write a program that reads an integer entered by the user and displays it in octal (base 8).

```

#include<stdio.h>

int main ()

{

```

```

int num,n1,n2,n3,n4,n5,d1,d2,d3,d4;

printf("Enter a number between 0 and 32767 : ");
scanf("%d",&num);

n1 = num%8;
d1 = num/8;

n2 = d1%8;
d2 = d1/8;

n3 = d2%8;
d3 = d2/8;

n4 = d3%8;
d4 = d3/8;

n5 = d4%8;

printf ("In octal, your number is : %d%d%d%d",n5,n4,n3,n2,n1);

return 0;
}

```

Task - 5

Rewrite the Program: Computing a UPC Check Digit from Lecture 04, so that the user enters 11 digits at one time, instead of entering one digit, then five digits, and then another five digits.

```

#include<stdio.h>

int main ()

{
    long long int k,d1,r1,d2,r2,d3,r3,d4,r4,d5,r5,d6,r6,d7,r7,d8,r8,d9,r9,d10,r10,d11,m,m2;

    int s1,s2,final_remin,check_digit;

    printf("\n\n\tEnter the first 11 digits of a UPC : ");
    scanf("%lld",&k);

    d1 = k/10000000000;
    r1 = k%10000000000;

    d2 = r1/1000000000;
    r2 = r1%1000000000;

    d3 = r2/100000000;
    r3 = r2%100000000;

    d4 = r3/10000000;
    r4 = r3%10000000;

    d5 = r4/1000000;
    r5 = r4%1000000;

    d6 = r5/100000;
    r6 = r5%100000;

    d7 = r6/10000;
    r7 = r6%10000;

    d8 = r7/1000;
    r8 = r7%1000;

    d9 = r8/100;
    r9 = r8%100;

    d10 = r9/10;
    r10 = r9%10;

    d11 = r10;

    s1 = d1+d3+d5+d7+d9+d11;
    s2 = d2+d4+d6+d8+d10;

    m = (s1*3)+s2;
    m2 = m - 1;
}

```

```

final_remin = m2%10;

check_digit = 9-final_remin;


printf("\n\n\tCheck digit : %d\n\n",check_digit);

}

```

Task - 6

European countries use a 13-digit code, known as a European Article Number (EAN) instead of the 12-digit Universal Product Code (UPC) found in North America. Each EAN ends with a check digit, just as a UPC does. The technique for calculating the check digit is also similar: (a) Add the 2nd,4th,6th,8th,10th,and 12th digits. (b) Add the 1st,3rd,5th,7th,9th,and 11th digits. (c) Multiply the first sum by 3 and add it to the second sum. (d) Subtract 1 from the total. (e) Compute the remainder when the adjusted total is divided

by 10. (f) Subtract the remainder from 9.

For example, consider Güllüoglu Turkish Delight Pistachio & Coconut, which has an EAN of 8691484260008. The first sum is $6+1+8+2+0+0 = 17$, and the second sum is $8+9+4+4+6+0 = 31$. Multiplying the first sum by 3 and adding the second yields 82. Subtracting 1 gives 81. The remainder on dividing by 10 is 1. When the remainder is subtracted from 9, the result is 8, which matches the last digit of the original code. Your job is to modify the upc.c program so that it calculates the check digit for an EAN. The

user will enter the first 12 digits of the EAN as a single number.

```

#include<stdio.h>

int main ()

{
    long long int k,d0,r0,d1,r1,d2,r2,d3,r3,d4,r4,d5,r5,d6,r6,d7,r7,d8,r8,d9,r9,d10,r10,d11,m,m2;

    int s1,s2,final_remin,check_digit;

    printf("\n\n\tEnter the first 12 digits of an EAN : ");
    scanf("%lld",&k);

    d0 = k/1000000000000;
    r0 = k%1000000000000;

    d1 = r0/100000000000;
    r1 = r0%100000000000;

    d2 = r1/10000000000;
    r2 = r1%10000000000;

    d3 = r2/1000000000;
    r3 = r2%1000000000;

    d4 = r3/100000000;
    r4 = r3%100000000;

    d5 = r4/10000000;
    r5 = r4%10000000;

    d6 = r5/1000000;
    r6 = r5%1000000;

    d7 = r6/100000;
    r7 = r6%100000;

    d8 = r7/10000;
    r8 = r7%10000;

    d9 = r8/1000;
    r9 = r8%1000;

    d10 = r9/100;
    r10 = r9%100;

    d11 = r10;

```

```

s1 = d1+d3+d5+d7+d9+d11;
s2 = d0+d2+d4+d6+d8+d10;

m = (s1*3)+s2;
m2 = m - 1;
final_remin = m2%10;

check_digit = 9-final_remin;

printf("\n\n\tCheck digit : %d\n\n",check_digit);

}

```

Task - 7

Write a program that stores two integers in two variables and then swaps the values of the two variables.

```

#include<stdio.h>

int main()
{
    int a,b,temp;

    printf ("input the first number   a = ");
    scanf ("%d",&a);

    printf ("input the second number  b = ");
    scanf ("%d",&b);

    temp = a;

    a = b;

    b = temp;

    printf("\n\na = %d\nb = %d\n\n",a,b);

    return 0;

}

```

Task - 8

Modify the program from the previous task so that it doesn't use any temporary variable.

```

#include <stdio.h>

int main ()
{
    int num_1,num_2;

    printf ("input the first number   num_1 = ");
    scanf ("%d",&num_1);

    printf ("input the second number  num_2 = ");
    scanf ("%d",&num_2);

    num_1 = num_1 + num_2;

    num_2 = num_1 - num_2;

    num_1 = num_1 - num_2;

    printf("\n\nnum_1 = %d\nnum_2 = %d\n\n",num_1,num_2);
}

```

```
    return 0;
}
```

Lab - 5 6/9/22

Task - 1

Students of International University of Tokyo (IUT) are famous for counting things. Anything you ask them, from flag stands, fields, departments, lakes, even leaves of trees, they can immediately tell you their count. The difficulty of such an answer depends on how many digits are there in that count. The more the difficulty, the greater the reward. Alice, a freshman at IUT, wants to know the reward she's going to get from answering questions. She knows the answers but she wants an automated system to find out the difficulty of each answer. Being her friend, the responsibility has fallen on your hands to help her. Write a program that calculates how many digits a number contains.

```
#include<stdio.h>

int main ()
{
    int num,det;

    printf("Enter a number : ");
    scanf("%d",&num);

    det = num/10;

    if (det==0)
    {
        printf("The number %d has 1 digit",num);
    }

    else if (det>0 && det<10)
    {
        printf("The number %d has 2 digits",num);
    }

    else if (det>=10 && det<100)
    {
        printf("The number %d has 3 digits",num);
    }

    else if (det>=100 && det<1000)
    {
        printf("The number %d has 4 digits",num);
    }

    else
    {
        printf("please input a number between 0 ~ 9999\n\n");
    }

    return 0;
}
```

Task - 2

Write a program that asks the user for a 24-hour time, then displays the time in 12-hour form

```
#include<stdio.h>

int main()

{
    int h,m,hr;

    printf("Enter a 24 hr time : ");
    scanf( "%d:%d",&h,&m);

    if (h>12 && h<=24)
    {
        hr = h - 12;
        printf("Equivalent 12-hour time : %d:%d PM\n\n",hr,m);
    }

    if (h==0)
    {
        printf("Equivalent 12-hour time : 12:%d AM\n\n",m);
    }

    else
    {
        printf("Equivalent 12-hour time : %d:%d AM\n\n",h,m);
    }

    return 0;

}
```

Task - 3

When stocks are sold or purchased through a broker, the broker's commission often depends upon the value of the stocks traded.

Suppose that a broker charges the amounts shown in the following table:

Under \$2,500	\$30 + 1.7%
\$2,500–\$6,250	\$56 + 0.66%
\$6,250–\$20,000	\$76 + 0.34%
\$20,000–\$50,000	\$100 + 0.22%
\$50,000–\$500,000	\$155 + 0.11%
Over \$500,000	\$255 + 0.09%

```
#include<stdio.h>

int main()
{
    float val,rate,num,p_s,rate_2;

    printf("Enter the number of shares : ");
    scanf("%f",&num);
    printf("Enter the price per share : ");
    scanf("%f",&p_s);

    val = num*p_s;

    // calculating original's values

    printf("\n\nThe original Broker...\n\n");

    if(val<529.42)
    {
        printf("Commission : $39");
    }

    else if(val>=529.42 && val<2500)
    {
        rate = 30 + ((val*1.7)/100);
        printf("commission : $%.2f",rate);
    }
}
```

```

else if(val>=2500 && val<6250)
{
    rate = 56 + ((val*0.66)/100);
    printf("commission : %.2f",rate);
}

else if(val>=6250 && val<20000)
{
    rate = 76 + ((val*0.34)/100);
    printf("commission : %.2f",rate);
}

else if(val>=20000 && val<50000)
{
    rate = 100 + ((val*0.22)/100);
    printf("commission : %.2f",rate);
}

else if(val>=50000 && val<500000)
{
    rate = 155 + ((val*0.11)/100);
    printf("commission : %.2f",rate);
}

else if(val>=500000)
{
    rate = 255 + ((val*0.09)/100);
    printf("commission : %.2f",rate);
}

//now calculating the rival's values
printf("\n\n\nThe rival broker...\n\n");

if (num<2000)
{
    rate_2 = 33 + ((p_s*.03f)*num);
    printf("commission : %.2f",rate_2);
}

else if (num>=2000)
{
    rate_2 = 33 + ((p_s*0.02f)*num);
    printf("commission : %.2f",rate_2);
}

printf("\n\n\n");

return 0;
}

```

Task - 4

4. **Beaufort Scale:**

Wind force is estimated by Beaufort Scale. Here's a simplified version of the scale:

Speed (knots)	Description
Less than 1	Calm
1-3	Light air
4-27	Breeze
28-47	Gale
48-63	Storm
Above 63	Hurricane

Write a program that asks the user to enter a wind speed (in knots), then displays the corresponding description.

```
#include<stdio.h>

int main()
{
    float s;

    printf("Enter wind speed in knots : ");
    scanf("%f",&s);

    if (s<1)
    {
        printf("Calm");
    }

    else if (s>=1 && s<=3)
    {
        printf("Light air");
    }

    else if (s>=4 && s<=27)
    {
        printf("Breeze");
    }

    else if (s>=28 && s<=47)
    {
        printf("Gale");
    }

    else if (s>=48 && s<=63)
    {
        printf("Storm");
    }

    else
    {
        printf("Hurricane");
    }

    return 0;
}
```

Task - 5

Modify the upc.c program from the previous lab so that it checks whether a UPC is valid. After the user enters a UPC, the program will display either VALID or NOT VALID

```

#include<stdio.h>

int main ()

{
    long long int k,d1,r1,d2,r2,d3,r3,d4,r4,d5,r5,d6,r6,d7,r7,d8,r8,d9,r9,d10,r10,d11,m,m2,d0,r0;

    int s1,s2,final_remin,check_digit;

    printf("\n\n\tEnter UPC  : ");
    scanf("%lld",&k);

    d0 = k/100000000000;
    r0 = k%100000000000;

    d1 = r0/10000000000;
    r1 = r0%10000000000;

    d2 = r1/1000000000;
    r2 = r1%1000000000;

    d3 = r2/100000000;
    r3 = r2%100000000;

    d4 = r3/10000000;
    r4 = r3%10000000;

    d5 = r4/1000000;
    r5 = r4%1000000;

    d6 = r5/100000;
    r6 = r5%100000;

    d7 = r6/10000;
    r7 = r6%10000;

    d8 = r7/1000;
    r8 = r7%1000;

    d9 = r8/100;
    r9 = r8%100;

    d10 = r9/10;
    r10 = r9%10;

    d11 = r10;

    s1 = d0+d2+d4+d6+d8+d10;
    s2 = d1+d3+d5+d7+d9;

    m = (s1*3)+s2;
    m2 = m - 1;
    final_remin = m2%10;

    check_digit = 9-final_remin;

    if (d11==check_digit)
    {
        printf("\n\tVALID");
    }

    else
    {
        printf("\n\tNOT VALID");
    }

    printf ("\n\n\n\n");
}

```

Task - 6

6. Tax Season:

In one state, single residents are subject to the following income tax:

<i>Income</i>	<i>Amount of Tax</i>
Not over \$750	1% of income
\$750 - \$2250	\$7.50 plus 2% of amount over \$750
\$2250 - \$3750	\$37.50 plus 3% of amount over \$2250
\$3750 - \$5250	\$82.50 plus 4% of amount over \$3750
\$5250 - \$7000	\$142.50 plus 5% of amount over \$5250
Over \$7000	\$230.00 plus 6% of amount over \$7000

Write a program that asks the user to enter the amount of taxable income, then displays the tax due.

```
#include<stdio.h>

int main()
{
    float in, tax, ret, least ;

    printf("Input your Taxable income : ");
    scanf("%f",&in);

    if (in<=750)
    {
        least=750;
        ret = in - least;
        tax = in * 0.01f;
        printf("Tax due is %.2f",tax);
    }

    else if (in>750 && in<=2250)
    {
        least = 750;
        ret = in - least;
        tax = 7.50f + (ret * 0.02f);
        printf("Tax due is %.2f",tax);
    }

    else if (in>2250 && in<=3750)
    {
        least = 2250;
        ret = in - least;
        tax = 37.50f + (ret * 0.03f);
        printf("Tax due is %.2f",tax);
    }

    else if (in>3750 && in<=5250)
    {
        least = 3750;
        ret = in - least;
        tax = 82.50f + (ret * 0.04f);
        printf("Tax due is %.2f",tax);
    }

    else if (in>5250 && in<=7000)
    {
        least = 5250;
        ret = in - least;
        tax = 142.5f + (ret * 0.05f);
        printf("Tax due is %.2f",tax);
    }

    else
    {
        least = 7000;
        ret = in - least;
```

```

        tax = 230.0f + (ret * 0.06f);
        printf("Tax due is %.2f",tax);
    }

    printf("\n\n\n\n");

    return 0;
}

```

Task - 7

Write a program that finds the largest and smallest of four integers entered by the user:

```

#include<stdio.h>

int main()

{
    int a,b,c,d,big_a,small_a,big_b,small_b,bigg,smalll;

    printf("Enter four integers : ");
    scanf("%d %d %d %d",&a,&b,&c,&d);

    if (a>b)
    {
        big_a = a;
        small_a = b;
    }
    else
    {
        big_a = b;
        small_a = a;
    }

    if (c>d)
    {
        big_b = c;
        small_b = d;
    }
    else
    {
        big_b = d;
        small_b = c;
    }

    if (big_a > big_b)
    {
        bigg = big_a;
    }
    else
    {
        bigg = big_b;
    }

    if (small_a < small_b)
    {
        smalll = small_a;
    }
    else
    {
        smalll = small_b;
    }

    printf("\n\n\nBiggest : %d\nSmallest : %d\n\n\n",bigg,smalll);

    return 0;
}

```

Task - 8

8. Welcome Aboard:

The following table shows the daily flights from one city to another:

<i>Departure time</i>	<i>Arrival time</i>
8:00 a.m.	10:16 a.m.
9:43 a.m.	11:52 a.m.
11:19 a.m.	1:31 p.m.
12:47 p.m.	3:00 p.m.
2:00 p.m.	4:08 p.m.
3:45 p.m.	5:55 p.m.
7:00 p.m.	9:20 p.m.
9:45 p.m.	11:58 p.m.

Write a program that asks the user to enter a time (expressed in hours and minutes, using the 24-hour clock). The program then displays the departure and arrival times for the flight whose departure time is closest to that entered by the user.

```
#include <stdio.h>

int main()
{
    int hr, min, conv_min, am_8, am_9_43, am_11_19, pm_12_47, pm_2, pm_3_45, pm_7, pm_9_45, am_12;

    printf("Enter a 24-hr time (hh:mm): ");
    scanf("%d:%d", &hr, &min);

    conv_min = hr * 60 + min;
    am_8 = 8 * 60;
    am_9_43 = 9 * 60 + 43;
    am_11_19 = 11 * 60 + 19;
    pm_12_47 = 12 * 60 + 47;
    pm_2 = 14 * 60;
    pm_3_45 = 15 * 60 + 45;
    pm_7 = 19 * 60;
    pm_9_45 = 21 * 60 + 45;
    am_12 = 23 * 60 + 59;

    if (conv_min >= 24 * 60 || conv_min <= 0)
        printf("Invalid hh:mm entry");
    else {
        printf("Closest departure time is: ");

        if (conv_min >= 1 && conv_min < am_8)
        {
            printf("8:00 am, arriving at 10:15 am");
        }

        if (conv_min >= am_8 && conv_min < am_9_43)
        {
            if (am_9_43 - conv_min < conv_min - am_8)
            {
                printf("9:43 am, arriving at 11:52 am");
            }
            else
            {
                printf("8:00 am, arriving at 10:15 am");
            }
        }

        if (conv_min >= am_9_43 && conv_min < am_11_19)
        {
            if (am_11_19 - conv_min < conv_min - am_9_43)
            {
                printf("11:19 am, arriving at 1:30 pm");
            }
            else
            {
                printf("9:43 am, arriving at 11:52 am");
            }
        }
    }
}
```

```

    }

    if (conv_min >= am_11_19 && conv_min < pm_12_47)
    {
        if (pm_12_47 - conv_min < conv_min - am_11_19)
        {
            printf("12:47 pm, arriving at 3:00 pm");
        }
        else
        {
            printf("11:19 am, arriving at 1:30 pm");
        }
    }

    if (conv_min >= pm_12_47 && conv_min < pm_2)
    {
        if (pm_2 - conv_min < conv_min - pm_12_47)
        {
            printf("2:00 pm, arriving at 4:08 pm");
        }
        else
        {
            printf("12:47 pm, arriving at 3:00 pm");
        }
    }

    if (conv_min >= pm_2 && conv_min < pm_3_45)
    {
        if (pm_3_45 - conv_min < conv_min - pm_2)
        {
            printf("3:45 pm, arriving at 5:55 pm");
        }
        else
        {
            printf("2:00 pm, arriving at 4:08 pm");
        }
    }

    if (conv_min >= pm_3_45 && conv_min < pm_7)
    {
        if (pm_7 - conv_min < conv_min - pm_3_45)
        {
            printf("7:00 pm, arriving at 9:20 pm");
        }
        else
        {
            printf("3:45 pm, arriving at 5:55 pm");
        }
    }

    if (conv_min >= pm_7 && conv_min < pm_9_45)
    {
        if (pm_9_45 - conv_min < conv_min - pm_7)
        {
            printf("9:45 pm, arriving at 11:58 pm");
        }
        else
        {
            printf("7:00 pm, arriving at 9:20 pm");
        }
    }

    if (conv_min >= pm_9_45 && conv_min <= am_12)
    {
        printf("9:45 pm, arriving 11:58 pm");
    }
}

printf("\n\n");

return 0;
}

```

Task - 9

Write a program that prompts the user to enter two dates and then indicates which date comes earlier on the calendar.

```
#include<stdio.h>
```

```

int main()

{
    int y1,y2,m1,m2,d1,d2,prev;

    printf("Enter first date (mm/dd/yy) : ");
    scanf( "%d/%d/%d", &m1,&d1,&y1);

    printf("\n\nEnter second date (mm/dd/yy) : ");
    scanf( "%d/%d/%d", &m2,&d2,&y2);

    if (y1>y2)
    {
        prev = 1;
    }

    else if (y1==y2)
    {
        if (m1>m2)
        {
            prev = 1;
        }

        else if (m1==m2)
        {
            if (d1>d2)
            {
                prev = 1;
            }
            else if (d1==d2)
            {
                printf("same date");
            }
            else
            {
                prev = 0;
            }
        }

        else
        {
            prev = 0;
        }
    }
    else
    {
        prev = 0;
    }

    printf("\n\n\n");

    if (prev == 1)
    {
        printf("%d/%d/%d is earlier than %d/%d/%d",m2,d2,y2,m1,d1,y1);
    }
    else if (prev == 0)
    {
        printf("%d/%d/%d is earlier than %d/%d/%d",m1,d1,y1,m2,d2,y2);
    }

    else
    {
        printf("wrong input");
    }

    printf("\n\n\n");

    return 0;
}

```

Task - 10

Using the switch statement, write a program that converts a numerical grade into a letter grade.

```

#include <stdio.h>

int main()

{

```

```

int grade;

printf("\nEnter a numerical grade: ");
scanf("%d", &grade);

if (grade < 0 || grade > 100)
{
    printf("Error !!! \n\n");
}

grade = grade - (grade % 10);

printf("Letter grade: ");

switch (grade) {
    case 100: case 90:
        printf("A");
        break;
    case 80: printf("B");
        break;
    case 70: printf("C");
        break;
    case 60: printf("D");
        break;
    case 50:
    case 40:
    case 30:
    case 20:
    case 10:
    case 0: printf("F");
        break;
}

printf("\n\n");

return 0;
}

```

Task - 11

Write a program that asks the user for a two-digit number, then prints the English word for the number.

```

#include <stdio.h>

int main()
{
    int number, a, b;

    printf("\nEnter a two-digit number: ");
    scanf("%d", &number);

    if (number < 10 || number > 99) {
        printf("The number did not contain two digits\n\n");
        return 0;
    }

    a = number - (number % 10);
    b = number % 10;

    printf("You entered the number ");

    if (a == 10) {
        switch (b) {
            case 0: printf("ten");
                break;
            case 1: printf("eleven");
                break;
            case 2: printf("twelve");
                break;
            case 3: printf("thirteen");
                break;
            case 4: printf("fourteen");
                break;
            case 5: printf("fifteen");
                break;
            case 6: printf("sixteen");
                break;
            case 7: printf("seventeen");

```



```

        break;
    case 8: printf("eighteen");
        break;
    case 9: printf("nineteen");
        break;
    }
    printf(".\n\n");
    return 0;
}

switch (a) {
    case 20: printf("twenty");
        break;
    case 30: printf("thirty");
        break;
    case 40: printf("forty");
        break;
    case 50: printf("fifty");
        break;
    case 60: printf("sixty");
        break;
    case 70: printf("seventy");
        break;
    case 80: printf("eighty");
        break;
    case 90: printf("ninety");
        break;
}

switch (b) {
    case 1: printf("-one");
        break;
    case 2: printf("-two");
        break;
    case 3: printf("-three");
        break;
    case 4: printf("-four");
        break;
    case 5: printf("-five");
        break;
    case 6: printf("-six");
        break;
    case 7: printf("-seven");
        break;
    case 8: printf("-eight");
        break;
    case 9: printf("-nine");
        break;
}

printf(".\n\n");

return 0;
}

```

Lab - 6 20/9/22

Task - 1

Write a program that finds the largest in a series of numbers entered by the user. The program must prompt the user to enter numbers one by one. When the user enters 0 or a negative number, the program must display the largest nonnegative number entered.

```

#include<stdio.h>

int main()
{
    float a,temp;
    temp = 0;

    while(1)

```

```

{
    printf("Enter a number : ");
    scanf("%f",&a);

    if(a>=temp)
    {
        temp = a;
    }

    if(a<=0)
    {
        break;
    }
}

printf("The largest number entered was %f",temp);

return 0;
}

```

Task - 2

D Finder: Write a program that asks the user to enter two integers, then calculates and displays their greatest common divisor (GCD).

```

#include<stdio.h>
int main()

{
    int p,q,i,gcd;

    printf("Enter two digits : ");
    scanf("%d %d",&p,&q);

    for(i=1;i<=p && i<=q;++i)
    {
        if(p%i==0 && q%i==0)
        {
            gcd = i;
        }
    }

    printf("Greatest common division : %d",gcd);

    return 0;
}

```

Task - 3

Write a program that asks the user to enter a fraction, then reduces the fraction to lowest terms.

```

#include<stdio.h>

int main()
{
    int lob,hor,i;

    printf("Enter a fraction : ");
    scanf("%d/%d",&lob,&hor);

    for(i=1;i<=lob&&i<=hor;++i)
    {
        if(lob%i==0&&hor%i==0)
        {
            lob = lob/i;
            hor = hor/i;

            i = 1;
        }
    }
}

```

```
printf("in lowest terms : %d/%d",lob,hor);

return 0;

}
```

Task - 4

4. Diamond Hands 💎:

When stocks are sold or purchased through a broker, the broker's commission often depends upon the value of the stocks traded.

Suppose that a broker charges the amounts shown in the following table:

<i>Transaction size</i>	<i>Commission rate</i>
Under \$2,500	\$30 + 1.7%
\$2,500–\$6,250	\$56 + 0.66%
\$6,250–\$20,000	\$76 + 0.34%
\$20,000–\$50,000	\$100 + 0.22%
\$50,000–\$500,000	\$155 + 0.11%
Over \$500,000	\$255 + 0.09%

The minimum charge is **\$39**

```
#include<stdio.h>

int main()
{
    double vtr,cmns;

    while(1)
    {
        printf("\n\nEnter value of trade : ");
        scanf("%lf",&vtr);

        if(vtr<2500 && vtr>0)
        {
            cmns = (30 + (vtr*0.017));
        }
        else if(vtr>=2500 && vtr<6250)
        {
            cmns = (56 + (vtr*0.0066));
        }
        else if(vtr>=6350 && vtr<20000)
        {
            cmns = (76 + (vtr*0.0034));
        }
        else if(vtr>=20000 && vtr<50000)
        {
            cmns = (100 + (vtr*0.0022));
        }
        else if(vtr>=50000 && vtr<500000)
        {
            cmns = (155 + (vtr*0.0011));
        }
        else if (vtr>=500000)
        {
            cmns = (255 + (vtr*0.0009));
        }

        if(vtr>0)
        {
            printf("\nCommission : %.2lf",cmns);
        }
    }
}
```

```

    }
    else
    {
        break;
    }
}

return 0;
}

```

Task - 6

In Lab 5 - Selection Statements (compare_dates.c), you were asked to write a program that determines which of two dates comes earlier on the calendar. Generalize the program so that the user may enter any number of dates. The user will enter 0/0/0 to indicate that no more dates will be entered.

```

#include <stdio.h>

int main()
{
    int m1, d1, y1, m2, d2, y2, earliest, latest;

    m1 = 0;
    d1 = 0;
    y1 = 0;

    printf("Enter a date (mm/dd/yy): ");
    scanf("%d/%d/%d", &m2, &d2, &y2);

    latest = (m2 * 30) + d2 + (y2 * 365);

    if (latest == 0) {
        printf("\n");
        return 0;
    }

    earliest = latest;

    while (latest != 0)
    {
        if (latest <= earliest)
        {
            m1 = m2;
            d1 = d2;
            y1 = y2;
            earliest = latest;
        }

        printf("Enter a date (mm/dd/yy): ");
        scanf("%d/%d/%d", &m2, &d2, &y2);
        latest = (m2 * 30) + d2 + (y2 * 365);
    }

    printf("\n%d/%d/%.2d is the earliest date.\n\n", m1, d1, y1);

    return 0;
}

```